

# ADITYA RAMAKRISHNAN

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## Education

### Michigan State University, College of Engineering

East Lansing, MI

*Bachelor of Science in Computer Science | Concentration in Artificial Intelligence*

May 2028

- GPA: 3.85/4.0 | Dean's List: Fall 2024, Spring 2025, Fall 2025
- Relevant Coursework: Data Structures and Algorithms, Computer Organization and Architecture, Object Oriented Software Design, Info Management and the Cloud, Intro to Machine Learning, AI Agents, Linear Algebra

## Experience

### Capital One

June 2026 - Aug 2026

*Software Engineer Intern - Business Risk Operations*

Plano, Texas

- Built an end-to-end **RAG** pipeline indexing **3,700+** regulatory requirements in **Chroma**, using semantic retrieval and **GPT-OSS-120B** to generate compliance questions, reducing requirement applicability mapping time by **85%**
- Engineered an adapter pipeline to ingest business process data from **Snowflake** into the compliance questionnaire platform, enabling automated population of questionnaire answers and saving **2,200+** hours annually
- Designed a **FastAPI** validation service for AI-generated compliance questions, using a human-in-the-loop framework to flag low-quality questions and escalate them for review, improving question generation reliability by **70%**

### Humana

May 2025 - Dec 2025

*Software Engineer Intern - Enterprise AI*

Remote

- Developed a GenAI **REST API** service using **FastAPI** and **Python** to generate custom summaries of healthcare documents via Humana's **Large Language Model (LLM) Gateway**, streamlining analysis for **10+** teams
- Implemented a **Pytest** framework to validate a GenAI summarization API (**85%** code coverage), containerized the service with **Docker**, and supported deployment to **Azure Kubernetes Service (AKS)** for scalable delivery
- Created a **FastAPI** service leveraging **Microsoft Azure's Text-to-Speech** to convert SVP meeting transcripts into synthetic-voice podcasts, improving accessibility to executive communication for **10,000+** employees
- Built backend for ScrumMate, a GenAI stand-up reporting tool at Humana's IT Intern Hackathon, using **FastAPI**, **Python**, and **LLMs**, extracting blockers and key topics from **8+** meetings and reducing reporting time by **80%**

### 180 Degrees Consulting MSU

Feb 2025 - Present

*Project Manager/Technology Chair*

East Lansing, MI

- Conceptualized and created a proof-of-concept university-exclusive reselling platform for MSU students using **ReactJS** and **Firebase**, visualizing the market opportunity and projecting revenue of up to **\$98,000/yr**
- Led the development of a **ReactJS** website for a local trucking company in a team of **4**, deployed on **Vercel**, increasing quote request submissions by **25%** through improved UX design and optimized CTAs

## Projects

### Candor | LiteRT, Gemma 4 E2B, Kotlin

May 2026

- Engineered an edge-native mobile application at the Qualcomm x Google Developer Hackathon that leverages **LiteRT** to execute on-device **LLMs**, enabling recovering addicts to track substance abuse patterns with **100%** data privacy
- Optimized on-device resources by offloading intensive **LLM** inference to the **NPU** while delegating secondary heuristic calculations to the **GPU** via **LiteRT**, achieving full recovery insight generation in under **3** seconds
- Utilized a **Gemma 4 E2B** model to perform high-speed on-device inference, analyzing weeks of historical recovery logs to identify behavioral patterns and protective factors for users within a **32,000** token context window

### CareerRise Advance | ReactJS, FastAPI, SQLite

Nov 2025

- Built a full-stack web application at JPMorgan Chase's Code for Good Hackathon to visualize county-level economic mobility with a **React Leaflet** heatmap using **GeoJSON**, enabling faster analysis for nonprofit stakeholders
- Developed **FastAPI** backend to retrieve unemployment, education, rent burden, and income data for **29+** counties
- Stored socioeconomic indicators in a **SQLite** database and computed county-level economic mobility weighted sums

## Technical Skills

**Languages:** Java, Python, C++, HTML, CSS, JavaScript, SQL

**Developer Tools:** Git, VS Code, AWS, Azure DevOps, Linux, Docker, Kubernetes, Snowflake, Jenkins, Google Firebase

**Libraries/Frameworks:** FastAPI, ReactJS, LiteRT, OpenAI API, TailwindCSS, Pytest, Matplotlib